

BADAM RAJASEKHARA REDDY

Java Full Stack Engineer | Payments | Distributed Systems

Hyderabad, India | +91-6281298823 | rajasekharjava018@gmail.com

PROFESSIONAL SUMMARY

Java Full Stack Engineer with 4+ years of experience designing and developing enterprise-scale FinTech payment platforms across payment acquiring, payment switching, merchant onboarding, settlement, reconciliation, and real-time transaction processing. Strong expertise in **Java, Spring Boot, Microservices, REST APIs, Angular, JavaScript, HTML, CSS, Kafka, MongoDB, Redis, and ISO 8583**, with hands-on experience building high-throughput distributed systems, event-driven architectures, and secure payment infrastructure. Experienced in designing reusable platform components, payment SDKs, performance optimization, and cloud-native deployments using **AWS, Docker, Jenkins, and Nginx**. Proven ability to deliver scalable, production-ready solutions across the complete software development lifecycle.

CORE TECHNICAL SKILLS

- **Languages:** Java 8/17, JavaScript
 - **Backend & Architecture:** Spring Boot, Spring MVC, Spring Security, Microservices, REST APIs, Hibernate, JPA, Spring Cloud
 - **Cloud & DevOps:** AWS, Docker, Kubernetes, Jenkins, CI/CD, Nginx, Route 53, Git
 - **Distributed Systems:** Kafka, Event-Driven Architecture, Async Processing, CompletableFuture, Virtual Threads, Concurrency, Semaphores
 - **Payments:** Payment Acquiring, Payment Switching, ISO 8583, UPI, Settlement, Reconciliation, Refunds, Reversals, Payment Gateway
 - **Databases:** MongoDB, Redis, Elasticsearch
 - **Frontend:** Angular, TypeScript, RxJS, Angular Material, Module Federation
 - **Engineering Fundamentals:** Core Java, Collections, OOP
 - **Security:** JWT, AES, RSA, SSL/TLS
 - **Architecture:** System Design, Design Patterns, Caching, Load Balancing, Reverse Proxy
 - **Build & Tools:** Maven, Gradle
 - **Testing & Quality:** Junit
 - **Developer Productivity Tools:** Git, Codex, Claude
 - **Practices:** Agile/Scrum, SDLC, Code Reviews
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PROFESSIONAL EXPERIENCE

TOUCAN Payments India Private Limited — Software Engineer

May 2022 – Present

Product Context – Payment Gateway

Enterprise-scale FinTech payment platform powering merchant onboarding, payment acquiring, payment switching, real-time transaction processing, settlement, reconciliation, and operational monitoring. The platform handles 100K–150K+ daily card and UPI transactions across multiple payment networks and PSP integrations, enabling secure, scalable, and highly available payment infrastructure.

Key Contributions & Achievements:

- **Designed and developed 12+ Spring Boot microservices** powering merchant onboarding, payment processing, settlement, reconciliation, and reporting, supporting 100K–150K+ daily payment transactions across "distributed payment systems".
- Developed ISO 8583 message processing services for Visa, Mastercard, and RuPay, supporting multiple

payment transaction flows including Purchase, Refund, Reversal, UPI, COFT, and MOTO.

- **Designed Kafka-based asynchronous workflows** processing over **150K payment events** daily, enabling reliable retry orchestration, fault-tolerant transaction processing, and scalable inter-service communication.
 - **Engineered a high-throughput transaction processing pipeline** using Java Virtual Threads, asynchronous execution, and semaphore-based throttling, achieving sustained throughput of **8K TPS** while implementing backpressure to prevent downstream PSP overload during peak traffic.
 - **Engineered a reusable transaction processing framework** using Factory, Strategy, and Template Method design patterns to support Purchase, Refund, Reversal, UPI, COFT, and MOTO transaction lifecycles.
 - **Implemented retry strategies, timeout handling**, and failure recovery mechanisms, improving payment reliability and transaction success during intermittent PSP outages.
 - **Implemented idempotency and transaction state management** to prevent duplicate processing and ensure transaction consistency across distributed payment workflows.
 - **Designed and developed a reusable JavaScript Authentication SDK** for PCI DSS-certified merchant checkout applications, standardizing authentication and authorization workflows and reducing **merchant integration time by 80%**.
 - **Reduced dashboard latency by approximately 70%** through MongoDB aggregation optimization, query tuning, early filtering, projections, and compound indexing.
 - Implemented Redis caching for frequently accessed payment metadata, reducing API response times by 55% and lowering database load by 40%.
 - **Engineered high-performance reporting services** using Java Streams, batch processing, and multithreading, achieving a sustained throughput of **13K records/second** for CSV, Excel, and PDF report generation while optimizing JVM memory utilization.
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EDUCATION

Bachelor of Technology (B.Tech)

Lovely Professional University, Phagwara, Punjab | 2021 | 83.38%